BEING LOCALLY OF FINITE PRESENTATION IS FPPF LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing phrase 'Hence we win.' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The general criterion for a property of morphisms to be fppf local on the source. Same chapter, Lemma lemma-properties-morphisms-local-source, invoked at proof lines 7039-7040. Reduces the claim to four clauses.
- Zariski locality and preservation under composition of local finite presentation. Morphisms, Lemmas morphisms-lemma-locally-finite-presentation-characterize and morphisms-lemma-composition-finite-presentation, invoked at proof lines 7034-7038. Supplies clauses (1)-(3).
- Permanence of flat finite presentation along a surjective flat finitely presented map of affines. Same chapter, Lemma lemma-flat-finitely-presented-permanence-algebra, invoked at proof lines 7041-7042. Supplies clause (4).

Source reference. Original lemma 02KK: sources/stacks-descent/pinned-original-source.tex, source lines 4478–4478. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 036H: sources/stacks-descent/pinned-original-source.tex, source lines 6920–6920. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01TR: sources/stacks-chow/tags.txt, source lines 2384–2384. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01TQ: sources/stacks-chow/tags.txt, source lines 2383–2383. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is locally of finite presentation" is fppf local on the source.*

Proof. Being locally of finite presentation is Zariski local on the source and the target, see Morphisms, Lemma 01TQ. It is a property which is preserved under composition, see Morphisms, Lemma 01TR. This proves (1), (2) and (3) of Lemma 036H. The final condition (4) is Lemma 02KK. Hence we win. □